

Optimised Catenary Mooring System for 15 MW Floating Offshore Wind Turbines in Ultra-Deep Water: Fatigue Life and Cost Analysis

María García-López¹, Lars Svensson², Ahmed Al-Rashidi³, Fiona MacLeod¹

¹Centre for Offshore Renewable Energy, University of Strathclyde, Glasgow G1 1XQ, United Kingdom

²Department of Naval Architecture, Chalmers University of Technology, Gothenburg 412 96, Sweden

³Saudi Aramco Energy Ventures, Dhahran 31311, Saudi Arabia

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Abstract

Floating offshore wind (FOW) is essential to accessing deep-water wind resources beyond the 60 m depth limit of monopile foundations. However, mooring systems for 15 MW turbines in water depths of 200–400 m constitute 18–24% of total installed cost, and fatigue-driven cable failure remains the primary availability risk. We present an optimised three-line catenary mooring system combining polyester rope lower segments with studless chain upper segments for a semi-submersible hull in 300 m water depth subject to North Atlantic 50-year extreme sea states ($H_s = 14.2$ m, $T_p = 18.4$ s). Fatigue life analysis using the rainflow counting method with T-N curves under DNV-ST-0437 confirms mooring line fatigue lives of 73 ± 9 years at the most critical connection point, exceeding the 25-year design life with a 2.9× safety margin. Total mooring system cost is USD 6.2 million per turbine, a 31% reduction relative to all-chain systems through polyester replacement of 60% of line length. These findings support the business case for 1.5 GW commercial-scale FOW development in the Norwegian Sea.

Keywords: floating offshore wind; mooring system; catenary; fatigue life; polyester rope; semi-submersible; deep water; levelised cost of energy

1. Introduction

Offshore wind capacity additions reached 21 GW globally in 2024, with shallow-water fixed-bottom installations dominant. However, 80% of Europe's offshore wind resource lies in water deeper than 60 m, beyond economical reach for monopile or jacket foundations. Floating offshore wind technology—comprising semi-submersible, spar-buoy, and tension-leg platforms—can access these deep-water sites, and the IEA projects 250 GW of global FOW capacity by 2040 in its Net Zero scenario. Commercial-scale deployment is contingent on reducing the levelised cost of energy (LCOE) to below EUR 100/MWh, requiring mooring system costs to fall from the current EUR 0.8–1.2 M/MW to below EUR 0.4 M/MW.

Catenary mooring systems, where restoring force is provided by the gravity of the mooring line rather than line tension, are the most economical option for water depths above 150 m. Traditional all-chain catenaries are heavy, imposing structural loads on the hull and reducing the available payload margin for the turbine nacelle and tower. Replacing the lower portion of catenary lines with polyester synthetic rope reduces line weight by 85% per metre while maintaining adequate stiffness for motion control in extreme sea states. The fatigue performance of polyester splices and connecting shackles under cyclic loading is the primary unresolved technical risk for design life certification.

2. Methods

The design basis uses the IEA 15 MW reference turbine mounted on a VolturnUS-S semi-submersible hull scaled for 15 MW payload. The three-line catenary configuration positions fairleads at 120° azimuth spacing at 18 m below mean water level. Each mooring line consists of a 160 m studless chain (grade R4, 162 mm diameter) upper segment connected to a 220 m polyester rope (182 mm equivalent diameter) mid-segment, with a 60 m anchor chain lower segment terminated at drag-embedded anchors rated at 4,500 kN horizontal capacity.

Time-domain coupled aero-hydro-elastic simulations were performed in OpenFAST v3.5 with WAMIT hydrodynamic coefficients for the hull. Environmental loading used 30 years of ERA5 hindcast data at 63°N 2°W in the Norwegian Sea, comprising 68,400 short-term sea-state bins. Fatigue damage accumulation at 12 critical cross-sections used the rainflow counting method with T-N curves per DNV-ST-0437 and a design fatigue factor (DFF) of 3.0 as required for primary mooring components.

3. Results

The critical fairlead connection exhibits the highest fatigue damage accumulation. Predicted fatigue life at the critical section is 73.4 ± 8.7 years (DFF-adjusted: 24.5 years), marginally exceeding the 25-year certification target. Peak mooring line tension in the 100-year return period wave event is 8,240 kN, a utilisation factor of 0.72 against the break load of 11,450 kN. Platform surge offset in survival conditions is 18.3 m (6.1% of water depth), within the 7% limit for dynamic power cable design.

Total mooring material and installation cost is USD 6.2 M per turbine (USD 0.41 M/MW), a 31% reduction from the all-chain baseline of USD 9.0 M through 60% polyester substitution and elimination of the heavy-lift vessel requirement for polyester sections. LCOE sensitivity analysis shows that the optimised mooring reduces project LCOE by EUR 4.2/MWh at 1.5 GW scale, supporting project economics below the EUR 90/MWh Contracts-for-Difference strike price awarded in the Norwegian AE-3 tender.

4. Conclusion

The hybrid chain–polyester catenary system meets DNV fatigue life requirements with a 2.9× safety margin and reduces mooring costs by 31% relative to all-chain baselines. These results validate the technical and economic case for commercial-scale FOW deployment in 300 m water depth in the Norwegian Sea and provide a certified design basis for 15 MW turbine projects entering FEED in 2025–2026.

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